

Sarthi Platform: HIPAA-Compliant GCP Services Implementation Guide

Appendices to Enterprise Architecture Document v1.5

Appendix A: HIPAA-Compliant Compute Services Configuration

A.1 Cloud Run Services (Multi-Tenant Architecture)

Based on the enterprise architecture's microservices design, each service requires HIPAA-compliant configuration:

yaml

sarthi_frontend_service:

service_name: "sarthi-frontend"

configuration:

vpc_connector: "sarthi-vpc-connector"

ingress: "INGRESS_TRAFFIC_INTERNAL_LOAD_BALANCER"

cpu_boost: true

execution_environment: "gen2"

encryption:

at_rest: "Customer-managed encryption keys (CMEK)"

in_transit: "TLS 1.3"

service_account: "sarthi-frontend-sa@sarthi-healthcare-prod.iam.gserviceaccount.com"

environment_variables:

HIPAA_COMPLIANT: "true"

AUDIT_LOGGING: "enabled"

PHI_HANDLING: "restricted"

sarthi_api_gateway:

service_name: "sarthi-api-gateway"

configuration:

vpc_connector: "sarthi-vpc-connector"

ingress: "INGRESS_TRAFFIC_INTERNAL_ONLY"

min_instances: 3 *# From multi-region scaling requirements*

max_instances: 100

cpu_boost: true

execution_environment: "gen2"

encryption:

at_rest: "Customer-managed encryption keys (CMEK)"

in_transit: "TLS 1.3"

service_account: "sarthi-api-gateway-sa@sarthi-healthcare-prod.iam.gserviceaccount.com"

ai_agent_orchestrator:

service_name: "sarthi-ai-orchestrator"

configuration:

vpc_connector: "sarthi-vpc-connector"

ingress: "INGRESS_TRAFFIC_INTERNAL_ONLY"

min_instances: 2 *# From capability-based routing requirements*

max_instances: 50

memory: "4GB"

cpu: "2 vCPU"

timeout: "900s"

encryption:

at_rest: "Customer-managed encryption keys (CMEK)"

in_transit: "TLS 1.3"

service_account: "sarathi-ai-orchestrator-sa@sarathi-healthcare-prod.iam.gserviceaccount.com"

llm_gateway_service:

service_name: "sarathi-llm-gateway"

configuration:

vpc_connector: "sarathi-vpc-connector"

ingress: "INGRESS_TRAFFIC_INTERNAL_ONLY"

min_instances: 2 *# From resilient LLM Gateway design*

max_instances: 30

memory: "4GB"

cpu: "2 vCPU"

timeout: "600s"

encryption:

at_rest: "Customer-managed encryption keys (CMEK)"

in_transit: "TLS 1.3"

service_account: "sarathi-llm-gateway-sa@sarathi-healthcare-prod.iam.gserviceaccount.com"

A.2 GKE Autopilot Configuration (For AI Agent Processing)

Following the enterprise architecture's containerized AI agent framework:

yaml

```
gke_autopilot_cluster:
  cluster_name: "sarathi-ai-agents-cluster"
  location: "us-central1"
  release_channel: "RAPID"
  network_config:
    network: "projects/sarathi-healthcare-prod/global/networks/sarathi-vpc"
    subnetwork: "projects/sarathi-healthcare-prod/regions/us-central1/subnetworks/sarathi-private-subnet"
    enable_private_nodes: true
    enable_private_endpoint: true
    master_ipv4_cidr_block: "172.16.0.0/28"

  security_config:
    workload_identity: true
    shielded_nodes: true
    binary_authorization: true
    network_policy: true
    enable_pod_security_policy: true

  monitoring_config:
    enable_cloud_logging: true
    enable_cloud_monitoring: true
    enable_managed_prometheus: true

  encryption_config:
    database_encryption:
      state: "ENCRYPTED"
      key_name: "projects/sarathi-healthcare-prod/locations/us-central1/keyRings/sarathi-gke/cryptoKeys/cluster-key"
```

Appendix B: HIPAA-Compliant Database Services

B.1 Cloud SQL PostgreSQL (Primary Database)

Based on the multi-model database strategy from the enterprise architecture:

```
yaml
```

cloud_sql_instance:

instance_id: "sarthi-primary-db-prod"

database_version: "POSTGRES_15"

region: "us-central1"

tier: "db-custom-8-32768" *# From scalability requirements*

backup_configuration:

enabled: true

start_time: "02:00"

location: "us-central1"

point_in_time_recovery: true

transaction_log_retention_days: 7

backup_retention_settings:

retained_backups: 30

retained_transaction_log_days: 7

maintenance_window:

day: 7 *# Sunday*

hour: 3

update_track: "stable"

ip_configuration:

require_ssl: true

private_network: "projects/sarthi-healthcare-prod/global/networks/sarthi-vpc"

authorized_networks: [] *# No public access*

enable_private_path_for_google_cloud_services: true

database_flags:

Enhanced logging for HIPAA compliance

- name: "log_statement"

value: "all"

- name: "log_min_duration_statement"

value: "1000"

- name: "log_connections"

value: "on"

- name: "log_disconnections"

value: "on"

- name: "log_checkpoints"

value: "on"

- name: "log_lock_waits"

value: "on"

- name: "log_temp_files"

value: "0"

```
- name: "shared_preload_libraries"  
  value: "pg_stat_statements,pg_audit"
```

```
deletion_protection: true
```

```
disk_encryption_configuration:
```

```
  kms_key_name: "projects/sarathi-healthcare-prod/locations/us-central1/keyRings/sarathi-phi/cryptoKeys/database-key"
```

```
replica_configuration:
```

```
  # Cross-region read replica for disaster recovery
```

```
  replica_names: ["sarathi-primary-db-replica-us-east1"]
```

```
  replica_zones: ["us-east1-a"]
```

```
tenant_isolation_databases:
```

```
  # Separate databases per tenant for enhanced isolation
```

```
  tenant_database_naming: "sarathi_tenant_{tenant_id}"
```

```
  max_connections_per_tenant: 50
```

```
  connection_pooling: "pgbouncer"
```

B.2 Memorystore Redis (Distributed Caching)

Following the performance optimization requirements:

```
yaml
```

```
memorystore_redis:
  instance_id: "sarathi-cache-prod"
  location_id: "us-central1-a"
  alternative_location_id: "us-central1-b" # HA configuration
  memory_size_gb: 16 # From caching strategy requirements
  tier: "STANDARD_HA"
```

```
auth_enabled: true
transit_encryption_mode: "SERVER_AUTHENTICATION"
auth_string: "stored-in-secret-manager"
```

```
maintenance_policy:
  weekly_maintenance_window:
    day: "SUNDAY"
    start_time:
      hours: 3
      minutes: 0
```

```
redis_configs:
  maxmemory-policy: "allkeys-lru"
  notify-keyspace-events: "Ex"
```

```
labels:
  environment: "production"
  compliance: "hipaa"
  data_classification: "phi"
```

B.3 Firestore (Document Storage)

For the real-time features mentioned in the enterprise architecture:

```
yaml
```

```
firestore_database:  
  project: "sarathi-healthcare-prod"  
  location_id: "us-central1"  
  type: "FIRESTORE_NATIVE"  
  concurrency_mode: "OPTIMISTIC"  
  app_engine_integration_mode: "DISABLED"  
  delete_protection_state: "DELETE_PROTECTION_ENABLED"
```

```
security_rules:  
  # Tenant isolation rules  
  tenant_based_access: true  
  phi_field_restrictions: true  
  audit_trail_required: true
```

```
backup_configuration:  
  daily_backup: true  
  retention_period: "2160h" # 90 days
```

```
point_in_time_recovery:  
  enabled: true  
  earliest_version_time: "24h"
```

Appendix C: HIPAA-Compliant Storage Services

C.1 Cloud Storage (PHI Data Storage)

Based on the document processing and AI agent requirements:

```
yaml
```

phi_storage_buckets:

primary_phi_bucket:

name: "sarhi-phi-storage-prod"

location: "US-CENTRAL1"

storage_class: "STANDARD"

uniform_bucket_level_access: true

public_access_prevention: "enforced"

encryption:

default_kms_key_name: "projects/sarhi-healthcare-prod/locations/us-central1/keyRings/sarhi-phi/cryptoKeys/stor"

retention_policy:

retention_period: 315360000 # 10 years (HIPAA requirement)

is_locked: true

lifecycle_rules:

- condition:

age: 90

action:

type: "SetStorageClass"

storage_class: "NEARLINE"

- condition:

age: 365

action:

type: "SetStorageClass"

storage_class: "COLDLINE"

- condition:

age: 2555 # 7 years

action:

type: "SetStorageClass"

storage_class: "ARCHIVE"

logging:

log_bucket: "sarhi-audit-logs-prod"

log_object_prefix: "phi-access/"

notification:

topic: "projects/sarhi-healthcare-prod/topics/phi-bucket-notifications"

event_types:

- "OBJECT_FINALIZE"

- "OBJECT_DELETE"

- "OBJECT_METADATA_UPDATE"

document_processing_bucket:

name: "sarathi-documents-processing-prod"

location: "US-CENTRAL1"

storage_class: "STANDARD"

uniform_bucket_level_access: true

public_access_prevention: "enforced"

encryption:

default_kms_key_name: "projects/sarathi-healthcare-prod/locations/us-central1/keyRings/sarathi-phi/cryptoKeys/stor"

lifecycle_rules:

- condition:

age: 30 # Documents processed within 30 days

action:

type: "Delete"

tenant_isolated_buckets:

naming_convention: "sarathi-tenant-{tenant_id}-{environment}"

per_tenant_encryption: true

cross_tenant_access_prevention: "enforced"

C.2 Secret Manager (Credentials & Keys)

Following the security architecture requirements:

yaml

```
secret_manager_configuration:
  secrets:
    database_credentials:
      secret_id: "sarathi-db-credentials"
    replication:
      user_managed:
        replicas:
          - location: "us-central1"
          - location: "us-east1"
      encryption:
        kms_key_name: "projects/sarathi-healthcare-prod/locations/us-central1/keyRings/sarathi-secrets/cryptoKeys/secret-"

    api_keys:
      secret_id: "sarathi-external-api-keys"
      replication:
        user_managed:
          replicas:
            - location: "us-central1"
            - location: "us-east1"
      rotation_policy:
        rotation_period: "2592000s" # 30 days
        next_rotation_time: "automated"

  tenant_encryption_keys:
    secret_id_pattern: "sarathi-tenant-{tenant_id}-key"
    per_tenant_isolation: true
    automatic_rotation: true

  topics:
    secret_version_add: "projects/sarathi-healthcare-prod/topics/secret-version-add"
    secret_version_destroy: "projects/sarathi-healthcare-prod/topics/secret-version-destroy"

  access_audit:
    logging_enabled: true
    alert_on_access: true
    unauthorized_access_detection: true
```

Appendix D: Encryption & Key Management (CMEK Strategy)

D.1 Cloud KMS Key Hierarchy

Based on the defense-in-depth security architecture:

yaml

kms_key_rings:

sarthi_phi_keyring:

name: "sarthi-phi"

location: "us-central1"

crypto_keys:

storage_encryption_key:

name: "storage-key"

purpose: "ENCRYPT_DECRYPT"

version_template:

algorithm: "GOOGLE_SYMMETRIC_ENCRYPTION"

protection_level: "SOFTWARE"

rotation_period: "7776000s" # 90 days

next_rotation_time: "auto-calculated"

labels:

environment: "production"

data_type: "phi"

compliance: "hipaa"

database_encryption_key:

name: "database-key"

purpose: "ENCRYPT_DECRYPT"

version_template:

algorithm: "GOOGLE_SYMMETRIC_ENCRYPTION"

protection_level: "HSM" # Hardware Security Module

rotation_period: "7776000s" # 90 days

labels:

environment: "production"

data_type: "phi"

compliance: "hipaa"

application_encryption_key:

name: "application-key"

purpose: "ENCRYPT_DECRYPT"

version_template:

algorithm: "GOOGLE_SYMMETRIC_ENCRYPTION"

protection_level: "SOFTWARE"

rotation_period: "15552000s" # 180 days

labels:

environment: "production"

data_type: "application"

sarthi_secrets_keyring:

name: "sarathi-secrets"

location: "us-central1"

crypto_keys:

secret_encryption_key:

name: "secret-key"

purpose: "ENCRYPT_DECRYPT"

version_template:

algorithm: "GOOGLE_SYMMETRIC_ENCRYPTION"

protection_level: "HSM"

rotation_period: "7776000s" # 90 days

tenant_specific_keyrings:

naming_pattern: "sarathi-tenant-{tenant_id}"

per_tenant_isolation: true

inheritance_from_master: false

dedicated_hsm: true # For enterprise customers

D.2 Encryption in Transit Configuration

yaml

tls_configuration:

minimum_version: "TLS 1.3"

cipher_suites:

- "TLS_AES_256_GCM_SHA384"
- "TLS_CHACHA20_POLY1305_SHA256"
- "TLS_AES_128_GCM_SHA256"

certificate_management:

provider: "Google-managed SSL certificates"

auto_renewal: true

internal_communication:

service_mesh: "Istio"

mutual_tls: "STRICT"

certificate_rotation: "automatic"

Appendix G: HIPAA Monitoring & Audit Logging

G.1 Cloud Logging Configuration

Based on the comprehensive audit logging requirements:

```yaml

audit\_logging:

cloud\_audit\_logs:

admin\_activity:

enabled: true

exempted\_members: []

data\_access:

enabled: true

log\_type: "DATA\_READ, DATA\_WRITE"

exempted\_members: []

custom\_log\_sinks:

phi\_access\_sink:

name: "sarathi-phi-access-logs"

destination: "bigquery.googleapis.com/projects/sarathi-healthcare-prod/datasets/phi\_audit\_logs"

filter: |

protoPayload.serviceName="storage.googleapis.com"

AND resource.labels.bucket\_name="sarathi-phi-storage-prod"

include\_children: true

authentication\_sink:

name: "sarhi-auth-logs"

destination: "logging.googleapis.com/projects/sarhi-healthcare-prod/logs/auth-events"

filter: |

```
protoPayload.methodName=("google.iam.admin.v1.CreateServiceAccountKey" OR
"google.iam.admin.v1.DeleteServiceAccountKey" OR
"google.identity.accesscontextmanager.v1.AccessPolicy")
```

database\_access\_sink:

name: "sarhi-database-logs"

destination: "bigquery.googleapis.com/projects/sarhi-healthcare-prod/datasets/database\_audit\_logs"

filter: |

```
resource.type="cloudsql_database"
AND protoPayload.serviceName="cloudsql.googleapis.com"
```

## G.2 Cloud Monitoring & Alerting

yaml

monitoring\_configuration:

uptime\_checks:

frontend\_uptime:

display\_name: "Sarathi Frontend Availability"

monitored\_resource:

type: "uptime\_url"

labels:

project\_id: "sarathi-healthcare-prod"

host: "app.sarathi.healthcare"

http\_check:

use\_ssl: true

path: "/health"

port: 443

timeout: "10s"

period: "60s"

api\_gateway\_uptime:

display\_name: "Sarathi API Gateway Availability"

monitored\_resource:

type: "uptime\_url"

labels:

project\_id: "sarathi-healthcare-prod"

host: "api.sarathi.healthcare"

http\_check:

use\_ssl: true

path: "/v1/health"

port: 443

timeout: "10s"

period: "60s"

alert\_policies:

hipaa\_breach\_detection:

display\_name: "HIPAA Potential Breach Alert"

conditions:

- display\_name: "Unauthorized PHI Access"

condition\_threshold:

filter: |

resource.type="gcs\_bucket"

AND resource.labels.bucket\_name="sarathi-phi-storage-prod"

AND protoPayload.authenticationInfo.principalEmail!~".\*@sarathi-healthcare-prod.iam.gserviceaccount.com"

comparison: "COMPARISON\_GREATER\_THAN"

threshold\_value: 0

duration: "60s"

notification\_channels:

- "projects/sarathi-healthcare-prod/notificationChannels/security-team"
- "projects/sarathi-healthcare-prod/notificationChannels/compliance-team"

alert\_strategy:

auto\_close: "1800s"

service\_availability:

display\_name: "Service Availability Alert"

conditions:

- display\_name: "Cloud Run Service Down"

condition\_threshold:

filter: |

resource.type="cloud\_run\_revision"

AND metric.type="run.googleapis.com/container/cpu/utilizations"

comparison: "COMPARISON\_LESS\_THAN"

threshold\_value: 0.01

duration: "300s"

notification\_channels:

- "projects/sarathi-healthcare-prod/notificationChannels/ops-team"

database\_performance:

display\_name: "Database Performance Alert"

conditions:

- display\_name: "High Connection Count"

condition\_threshold:

filter: |

resource.type="cloudsql\_database"

AND metric.type="cloudsql.googleapis.com/database/postgresql/num\_backends"

comparison: "COMPARISON\_GREATER\_THAN"

threshold\_value: 80

duration: "300s"

notification\_channels:

- "projects/sarathi-healthcare-prod/notificationChannels/dba-team"

## G.3 Security Command Center Configuration

yaml

```
security_command_center:
 findings_filter:
 high_severity_phi:
 filter: |
 category="SENSITIVE_DATA_EXPOSURE"
 AND state="ACTIVE"
 AND security_marks.marks.data_type="PHI"
 notification_config:
 pubsub_topic: "projects/sarathi-healthcare-prod/topics/security-findings"
 streaming_config:
 filter: "severity=\"HIGH\" OR severity=\"CRITICAL\""

custom_modules:
 phi_scanner:
 display_name: "PHI Data Scanner"
 enabled_state: "ENABLED"
 custom_config:
 predicate:
 expression: |
 resource.type == "gcs_bucket" &&
 resource.data.bucket.name.startsWith("sarathi-phi-")
 resource_selector:
 resource_types: ["google.storage.Bucket"]
```

## Appendix H: Backup & Disaster Recovery

### H.1 Database Backup Strategy

Following the enterprise architecture's multi-region resilience requirements:

```
yaml
```

backup\_configuration:

cloud\_sql\_backups:

automated\_backups:

enabled: true

start\_time: "02:00"

location: "us-central1"

point\_in\_time\_recovery: true

transaction\_log\_retention\_days: 7

backup\_retention\_settings:

retained\_backups: 30

retained\_transaction\_log\_days: 7

manual\_backups:

frequency: "weekly"

retention\_period: "1 year"

cross\_region\_replication: true

destinations:

- "us-east1"

- "us-west1"

cloud\_storage\_backups:

versioning:

enabled: true

max\_versions: 100

lifecycle\_management:

rules:

- condition:

  - age: 30

- action:

  - type: "SetStorageClass"

  - storage\_class: "NEARLINE"

- condition:

  - age: 90

- action:

  - type: "SetStorageClass"

  - storage\_class: "COLDLINE"

- condition:

  - age: 365

- action:

  - type: "SetStorageClass"

  - storage\_class: "ARCHIVE"

cross\_region\_replication:  
 enabled: true  
 destination\_bucket: "sarathi-phi-storage-backup-us-east1"  
 replication\_time:  
 status: "Enabled"  
metrics:  
 status: "Enabled"  
 event\_threshold:  
 event\_threshold: "15 minutes"

## H.2 Disaster Recovery Procedures

yaml

disaster\_recovery:

rpo\_targets: *# Recovery Point Objective*

database: "1 hour"

file\_storage: "4 hours"

application\_state: "15 minutes"

rto\_targets: *# Recovery Time Objective*

critical\_services: "1 hour"

non\_critical\_services: "4 hours"

full\_platform: "8 hours"

failover\_regions:

primary: "us-central1"

secondary: "us-east1"

tertiary: "us-west1"

automated\_failover:

health\_check\_failures: 3

failover\_timeout: "10 minutes"

traffic\_splitting:

gradual\_rollover: true

percentage\_steps: [10, 25, 50, 100]

manual\_procedures:

database\_restoration:

cross\_region\_replica\_promotion: "30 minutes"

point\_in\_time\_recovery: "2 hours"

full\_backup\_restoration: "6 hours"

application\_recovery:

cloud\_run\_redeployment: "15 minutes"

gke\_cluster\_recreation: "45 minutes"

dns\_failover: "5 minutes"

## Appendix I: Compliance Automation

### I.1 Cloud Security Command Center Integration

yaml

compliance\_automation:

security\_health\_analytics:

enabled\_detectors:

- "OPEN\_FIREWALL"
- "SSL\_NOT\_ENFORCED"
- "WEAK\_SSL\_POLICY"
- "PUBLIC\_BUCKET\_ACL"
- "PUBLIC\_DATASET"
- "WEAK\_KMS\_KEY"
- "ADMIN\_SERVICE\_ACCOUNT"
- "LEGACY\_AUTHORIZATION"

compliance\_monitoring:

hipaa\_controls:

- control\_id: "164.308(a)(1)(i)"  
description: "Security Officer Assignment"  
automated\_check: "IAM roles verification"  
frequency: "daily"
- control\_id: "164.308(a)(5)(ii)(D)"  
description: "Password Management"  
automated\_check: "Service account key rotation"  
frequency: "weekly"
- control\_id: "164.312(a)(1)"  
description: "Access Control"  
automated\_check: "IAM policy analysis"  
frequency: "daily"
- control\_id: "164.312(e)(1)"  
description: "Transmission Security"  
automated\_check: "TLS enforcement verification"  
frequency: "continuous"

automated\_remediation:

public\_bucket\_remediation:

trigger: "PUBLIC\_BUCKET\_ACL"  
action: "remove\_public\_access"  
approval\_required: false

weak\_ssl\_remediation:

trigger: "WEAK\_SSL\_POLICY"  
action: "update\_ssl\_policy"

approval\_required: true

firewall\_remediation:

trigger: "OPEN\_FIREWALL"

action: "restrict\_access"

approval\_required: true

## I.2 Policy-as-Code Implementation

yaml

```
policy_enforcement:
 organization_policies:
 compute_disable_serial_port:
 constraint: "compute.disableSerialPortAccess"
 enabled: true

 storage_uniform_bucket_access:
 constraint: "storage.uniformBucketLevelAccess"
 enabled: true

 sql_restrict_public_ip:
 constraint: "sql.restrictPublicIp"
 enabled: true

 iam_allowed_policy_member_domains:
 constraint: "iam.allowedPolicyMemberDomains"
 allowed_values: ["sarathi.healthcare"]

binary_authorization:
 policy:
 default_admission_rule:
 evaluation_mode: "REQUIRE_ATTESTATION"
 enforcement_mode: "ENFORCED_BLOCK_AND_AUDIT_LOG"
 require_attestations_by:
 - "projects/sarathi-healthcare-prod/attestors/prod-attestor"

 attestors:
 prod_attestor:
 name: "prod-attestor"
 description: "Production container attestor"
 user_owned_grafeas_note:
 note_reference: "projects/sarathi-healthcare-prod/notes/prod-note"
 public_keys:
 - pkix_public_key:
 public_key_pem: "${file("attestor-public-key.pem")}"
 signature_algorithm: "RSA_PSS_2048_SHA256"
```

## Appendix J: Multi-Environment Configuration

### J.1 Environment-Specific Variations

Based on the cost optimization strategy from the enterprise architecture:

yaml

environment\_configs:

development:

cost\_optimization: 90%

security\_level: "basic\_hipaa"

encryption:

database: "Google-managed keys"

storage: "Google-managed keys"

secrets: "Google-managed keys"

networking:

private\_google\_access: true

vpc\_peering: false

private\_service\_connect: false

monitoring:

retention\_days: 30

alerting: "development-team-only"

staging:

cost\_optimization: 50%

security\_level: "full\_hipaa"

encryption:

database: "Customer-managed keys"

storage: "Customer-managed keys"

secrets: "Customer-managed keys"

networking:

private\_google\_access: true

vpc\_peering: true

private\_service\_connect: true

monitoring:

retention\_days: 90

alerting: "staging-team-and-security"

production:

cost\_optimization: 0%

security\_level: "enterprise\_hipaa"

encryption:

database: "Customer-managed keys with HSM"

storage: "Customer-managed keys"

secrets: "Customer-managed keys with HSM"

networking:

private\_google\_access: true

vpc\_peering: true

private\_service\_connect: true

vpc\_service\_controls: true

```
monitoring:
 retention_days: 365
 alerting: "full-team-and-compliance"
```

## J.2 Terraform Environment Management

yaml

terraform\_structure:

modules:

- name: "hipaa-compute"  
path: "./modules/compute"  
description: "HIPAA-compliant compute services"
- name: "hipaa-database"  
path: "./modules/database"  
description: "HIPAA-compliant database services"
- name: "hipaa-storage"  
path: "./modules/storage"  
description: "HIPAA-compliant storage services"
- name: "hipaa-networking"  
path: "./modules/networking"  
description: "HIPAA-compliant VPC and networking"
- name: "hipaa-security"  
path: "./modules/security"  
description: "HIPAA security controls and monitoring"

environments:

- name: "dev"  
workspace: "sarathi-dev"  
tfvars: "./environments/dev/terraform.tfvars"
- name: "staging"  
workspace: "sarathi-staging"  
tfvars: "./environments/staging/terraform.tfvars"
- name: "prod"  
workspace: "sarathi-prod"  
tfvars: "./environments/prod/terraform.tfvars"

# Appendix K: Cost Optimization for HIPAA Environments

## K.1 Resource Optimization Strategy

Following the cost management requirements from the enterprise architecture:

yaml

cost\_optimization:

rightsizing:

cloud\_run\_services:

development:

min\_instances: 0

max\_instances: 10

cpu: "1 vCPU"

memory: "2GB"

staging:

min\_instances: 1

max\_instances: 25

cpu: "2 vCPU"

memory: "4GB"

production:

min\_instances: 3

max\_instances: 100

cpu: "2 vCPU"

memory: "4GB"

storage\_tiering:

immediate\_access: "STANDARD"

frequent\_access: "NEARLINE"

infrequent\_access: "COLDLINE"

archive\_storage: "ARCHIVE"

database\_optimization:

development:

tier: "db-f1-micro"

backup\_retention: 7

staging:

tier: "db-custom-2-8192"

backup\_retention: 30

production:

tier: "db-custom-8-32768"

backup\_retention: 365

read\_replicas: 2

scheduled\_operations:

dev\_environment\_shutdown:

```
schedule: "0 18 * * 1-5" # 6 PM weekdays
```

```
action: "scale_to_zero"
```

```
dev_environment_startup:
```

```
schedule: "0 8 * * 1-5" # 8 AM weekdays
```

```
action: "scale_to_normal"
```

## K.2 Cost Monitoring & Alerts

yaml

```
cost_monitoring:
```

```
 budgets:
```

```
 monthly_budget:
```

```
 amount: 50000 # $50K per month
```

```
 threshold_rules:
```

```
 - threshold_percent: 0.5
```

```
 spend_basis: "FORECASTED_SPEND"
```

```
 - threshold_percent: 0.8
```

```
 spend_basis: "CURRENT_SPEND"
```

```
 - threshold_percent: 1.0
```

```
 spend_basis: "CURRENT_SPEND"
```

```
cost_allocation:
```

```
 labels:
```

```
 environment: ["dev", "staging", "prod"]
```

```
 component: ["frontend", "backend", "database", "ai-agents"]
```

```
 tenant: ["shared", "tenant-specific"]
```

```
 compliance_level: ["basic", "hipaa", "enterprise"]
```

```
billing_exports:
```

```
 bigquery_dataset: "sarathi_billing_export"
```

```
 table_name: "gcp_billing_export"
```

```
 partition_field: "export_time"
```

## Implementation Notes

### Environment-Specific Variations

- **Development:** Reduced encryption requirements, simplified networking, cost-optimized resources
- **Staging:** Full HIPAA compliance for realistic testing, moderate cost optimization
- **Production:** Maximum security, all controls enabled, no cost compromises on security

## Compliance Validation

- All configurations include comprehensive audit logging
- Automated compliance scanning with Cloud Security Command Center
- Regular penetration testing and vulnerability assessments
- HIPAA compliance monitoring and automated alerting
- Policy-as-code enforcement with binary authorization

## Cost Optimization Balance

- Non-production environments use Google-managed encryption keys where permissible
- Development environments implement scheduled shutdown/startup for compute resources
- Staging environments use reduced backup retention and replica counts
- Production maintains full security posture regardless of cost implications

## Operational Excellence

- Infrastructure-as-code with Terraform for reproducible deployments
- Automated remediation for common security misconfigurations
- Comprehensive monitoring with business and technical metrics
- Disaster recovery procedures with defined RTO/RPO targets

This comprehensive implementation guide ensures that the Sarthi platform meets all HIPAA requirements while maintaining the scalability, reliability, and cost-effectiveness outlined in the Enterprise Architecture Document v1.5. The appendices provide actionable configurations for immediate implementation while supporting the long-term architectural vision.

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## Implementation Notes

### Environment-Specific Variations

- **Development:** Reduced encryption requirements, simplified networking
- **Staging:** Full HIPAA compliance for realistic testing
- **Production:** Maximum security, all controls enabled

### Compliance Validation

- All configurations include audit logging
- Automated compliance scanning with Cloud Security Command Center

- Regular penetration testing and vulnerability assessments
- HIPAA compliance monitoring and alerting

## **Cost Optimization**

- Non-production environments use standard encryption (Google-managed keys)
- Development environments use Cloud SQL Auth Proxy instead of private IP
- Staging uses smaller instance sizes with burst capabilities

This implementation guide ensures that the Sarthi platform meets all HIPAA requirements while maintaining the scalability, reliability, and cost-effectiveness outlined in the Enterprise Architecture Document v1.5.